How Do You Save the World's Chocolate Supply? You Reinvent Chocolate; Candy prices are up 44% since 2021, but two groups of innovators are striving to solve the growing cocoa-bean shortage

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FULL TEXT

Another day, another dramatic shortage of an economic building block, from memory chips to diesel fuel . Many of the factors weighing on those supply chains also affect food, and in particular that other necessity of modern life: chocolate. Candy prices are up 44% since 2021, according to data from the Bureau of Labor Statistics . But there's good news, as long as you're OK with eating chocolate grown in a vat—or made from something that isn't a cocoa bean.

Innovators are using techniques refined by the pharmaceutical industry to create "bioidentical" cocoa-bean cells, leading to the first bars made from experimental, lab-grown chocolate. Other companies are scaling up processes for roasting or fermenting agricultural waste products, to make them good-enough substitutes for the primary ingredients in chocolate. There are close to a dozen startups working on lab-grown chocolate and chocolate-flavored substitutes, in the U.S., Europe, Israel and Asia.

If these sound less than appetizing, experts in the chocolate-supply chain have a response: "I would challenge consumers to tell the difference," says Raphaël Felenbok, who advises some of the biggest candy makers and sits on the board of a chocolate-substitute startup.

You might already be a part of the industry's experimentation without realizing it. The Hershey Company switched the labeling on some favorites such as Almond Joy, Mr. Goodbar and Rolo to say "chocolate candy" rather than "milk chocolate," after incorporating fats other than milk fat and cocoa butter. "Every ingredient we use serves a specific purpose, whether it contributes to flavor, product performance, freshness or the overall eating experience," a company spokeswoman said.

Before lab-grown cocoa hits the market, the already available alternatives that contain no actual cocoa probably won't win any chocolatiering competitions. I tried a sample from a company called Voyage Foods, and I realized its strength is adding chocolaty vibes to cakes and confections, not replacing solid bars of milk chocolate, much less dark chocolate.



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Voyage Foods processes grapeseeds into a cocoa alternative. PHOTO: Andrew Spear for WSJ

But those bars of real chocolate might themselves be luxury items before long, since the world's supply of cocoa beans is in

crisis. The drivers include extreme weather—drought followed by flooding—and the economic crises that have impoverished growers and left them unable to upgrade their agricultural practices.

"In the cocoa space, there is a lot of activity," says Florian Schattenmann, chief technology officer at Cargill, the world's largest agricultural-commodities company. "Weather events can happen anytime," he adds.

There are also real questions about the sustainability of a system built on subsistence farming in some of the poorest regions on earth.

In 2024, the price for cocoa beans peaked at more than $12,000 for a metric ton , compared with the $2,000 to $3,000 a ton it had fetched in the decade prior. Prices have since stabilized to around $5,000 to $6,000 a ton, but could rise again soon.

The future: cell-cultured cocoa beans

When Chief Executive Michal Beressi Golomb first showed off an early batch of cultured cocoa-bean chocolate bars made by her startup Celleste, she didn't have enough for investors to taste. "Don't touch it," she recalls saying. "Those are my babies."

Those first bars—made in collaboration with Mondelez at the chocolate giant's R&D facility west of London—were so precious because the process was still very tricky.

The pharmaceutical industry creates drugs such as insulin, cancer treatments and some vaccines by growing substances within cells. They use ultrasterilized steel vessels, which adds to the high cost of the process.

The challenge in applying this technology to growing food instead of drugs is that the costs must come down by a factor of 10 or more, says Steven Lang. He is vice president of science and technology at California Cultured, a competitor to Celleste that is also working with cocoa-bean cells. He advocates shifting to cheaper, food-grade systems like plastic tubs.

California Cultured wants to grow bean cells high in flavanols and absent the heavy metals sometimes found in dark chocolate , to appeal to those who want to reap the health benefits of cocoa .

Celleste's initial goal is to manufacture using 1,000-liter bioreactors, each with a yearly output of a metric ton of cocoa. Typically, it takes about 5 acres of trees to make a ton; this would be done with just two beans. The company is currently developing its process using smaller vessels.

Both Golomb and Lang emphasize that what their companies are creating isn't a chocolate substitute. It's chocolate, full stop, indistinguishable from what comes from trees.

California Cultured hopes to take products to market by 2028, while Celleste is aiming for U.S. regulatory approval by then. Celleste plans to sell cocoa butter and powder as a supplemental supply to chocolate manufacturers and food companies besieged by rising traditional-bean prices.

The near term: roasted seeds, beans and grains

Food scientists like to point out that, sensorily, chocolate is more about the experience than the actual substance. Regulators in charge of food labeling tend to agree. Companies are allowed to replace some portion of actual chocolate in their treats with a close-enough substitute. This is known as a "silent swap," says Adam Maxwell, founder and CEO of Voyage Foods.

When I last caught up with his company, it plied me with artificial coffee. (It's better than it sounds .) Now, it's all in on cocoa substitutes. Voyage Foods' product is made from roasted and ground grapeseeds left over from the winemaking industry, blended with sunflower protein flour, palm and shea oils, sugar and natural flavors. Much of the company's manufacturing process mirrors that of traditional chocolate production.

Voyage Foods' new, 284,000-square-foot factory in Ohio is operational and able to produce 10,000 metric tons of cocoa substitute yearly, says Maxwell. Cargill is an investor, and the global distributor of its chocolate product.



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The production floor at Voyage Foods in Mason, Ohio. PHOTO: Andrew Spear for wsj

Other companies taking this approach include Win-Win in the U.K., which uses a mix of roasted grains in its cocoa substitute, and Prefer in Singapore, which uses a mix of seeds and grains. (For its part, Hershey says it has no active work in either cocoa substitutes or lab-grown cocoa.)

None of this is meant to trick the consumer, insists Maxwell. Rather, it's about keeping chocolate—or at least something that evokes its essence—affordable and available. "For me, it was always about, how can we have these beloved foods for

everyone, forever," he adds.

The world's ever-growing population of chocoholics will create a need for more than just these technologies, says Felenbok, who advises the Win-Win team. For instance, efforts are under way to grow cocoa trees in places to which they're not native, like Latin America.

But as the world gets warmer and more subject to extreme weather, chocolate substitutes will gain ground. If that troubles you, just think about all the vanilla-flavored treats you might enjoy, says Maxwell. Almost none of that taste comes from the actual plant.

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